
UNIT 11 MEASURES OF INEQUALITY

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11.0 OBJECTIVES

After going through this Unit, you will be able to:

- explain the various positive measures of inequality;
- discuss the computational device for construction of Gini index; and
- describe the normative measures of inequality propounded by Dalton, Atkinson, Sen and Theil.

11.1 INTRODUCTION

Improvement in well-being of the poor has been one of the important goals of economic policy and to a significant extent it is determined by the growth and distribution of its income. Distribution patterns have an important bearing on the relationship between average income and poverty levels. Extreme inequalities are economically wasteful. Further, income inequalities also interact with other life-chance inequalities. Hence reducing inequalities has become priority of public policy.

It therefore, becomes pertinent to measure income inequalities. Various measures have been developed over a period of time to study the level of inequalities in different situations. Broadly these measures can be put under two categories (i) positive measures, and (ii) normative measures. The measures which capture the inequality of income without value judgment about social well-being are known as positive measures. Range, quartile range, standard deviation, Gini ratio, Lorenz curve etc. are positive measures of inequality. On the other hand, the measures that essentially involve value judgement about social welfare are called normative measures. The index propounded by Dalton, Atkinson, Sen, Theil and Kakwani are normative measures. We shall discuss all these measures one by one in this unit. Gini Coefficient of Inequality and Lorenz Curve will receive particular attention, as they are very popular in literature.

11.2 POSITIVE MEASURES

If all values in a distribution are not equal, which means that there is dispersion in the distribution, hence, there exists inequality in the distribution. If a measure is developed to capture this non-equality in values without giving explicit consideration to its consequences with respect to social well-being or economic significance in a particular context, the measure is known as **positive**. It means that the measure is bothering about the fact whether it is measuring inequality of lengths of iron nails or incomes of wage earners in a village. Nevertheless, many of them are standard statistical measures and their social implications and/or social consequences can still be studied. Some of these measures can be arrived from normative approach as well.

Let us consider an income distribution x_i $i=1,2,\dots,N$ over N persons and with mean income μ . Let the relative share of total income with person i be designated as q_i , which is naturally given by $x_i/N\mu$. The cumulative share of total income with the persons not having more than x_i income can be given as Q_i . However, when we have an income having frequency more than one, the proportion of people with income x_i can be denoted by p_i and the cumulative proportion of people with income no less than x_i as P_i . In this case, obviously the relative share of total income would be given by $f_i x_i / N\mu$ where f_i denotes the frequency of occurrence of income x_i and $N = \sum f_i$. It will not be necessary to use two different subscripts for distinguishing the two cases for the purpose as the context would make it clear whether subscript i stands for a person with income x_i or for the group of persons with income x_i .

We have implicitly assumed that the data are arranged in the increasing (non-decreasing) order by magnitude of income so that symbolic representation is easy. The same could be accomplished by decreasing (non-increasing) order.

We intend to cover important measures along with their variants in this Unit. We shall also, albeit briefly, discuss their properties and weaknesses.

To recapitulate, we recount what we said above in a formal way

$$q_i = \frac{x_i}{N\mu} \text{ or } \frac{f_i x_i}{N\mu}$$

$$Q_i = \sum_{i=1}^i q_i,$$

$$p_i = \frac{1}{N} \text{ or } \frac{f_i}{N}$$

$$P_i = \sum_{i=1}^i p_i,$$

11.2.1 Relative Range

A measure of relative dispersion can be taken a measure of inequality. It is defined as the relative range by

$$RR_1 = \frac{Max_i x_i - Min_i x_i}{\mu} \quad (RR.1)$$

that is, the relative difference between the highest income and the lowest income. If income is equally distributed, then $RR_1 = 0$ and if one person received all the income, then RR_1 is maximum. If one wants to make the index lie in the interval between 0 and 1, one can define it as

$$RR_2 = \frac{Max_i x_i - Min_i x_i}{N\mu} \quad (RR.2)$$

which means it is the gap between the maximum share and the minimum share. That is,

$$RR_2 = Max_i q_i - Min_i q_i \quad (RR.3)$$

Though Cowell has suggested division of range by $Min_i x_i$, which does not serve, in our view, any purpose. Two other normalization or standardization procedures that make it unit-free and contain it in (0,1) interval are suggested below:

$$RR_3 = \frac{Max_i x_i - Min_i \bar{x}_i}{Max_i x_i} \quad (RR.4)$$

and

$$RR_4 = \frac{Max_i x_i - Min_i x_i}{Max_i x_i + Min_i x_i} \quad (RR.5)$$

The basic weaknesses of these range-based measures are that they are not based on all values and therefore they do not reflect the change in inequality if there is any transfer of income between two non-extreme recipients.

Instead of considering extreme values at either end, which may not be even known, some scholars have toyed with the idea of the ratio between the mean income of the highest fractile (percentile, quintile or decile) and that of the lowest counterpart. They term it as the extreme disparity ratio (EDR). Naturally, this ratio is not contained in the interval (0,1). This ratio is independent of μ as well. The measure will not reflect the transfer of income that does not involve the extreme fractiles.

11.2.2 Relative Inter-Quartile Range

Sometimes, extremism of the relative range is sought to be moderated by restricting the distribution between the 10th and 90th percentile or sometimes to interquartile range. Bowley (1937) suggested relative quartile deviation as the index of inequality:

$$B = \frac{x^{q^3} - x^{q^1}}{x^{q^3} + x^{q^1}} \quad (\text{B.1})$$

where x^{q^r} represents the income level which divides the population in r and $(4-r)$ quartiles. B is zero for degenerate distribution where everybody has the same income and unity if the lowest 75 per cent people have no income at all.

Though the extremes are moderated in comparison to the measure of range, it has an obvious weakness that the measure takes into account only 50 per cent of the distribution. Further, a transfer of income between two persons without causing either or both of them cross x^{q^1} or x^{q^3} would not change the measured level of inequality. Thus, the index suffers from all weaknesses of the earlier proposals except that of extremism. Its highest value reaches when the lowest 75 per cent people do not possess any income.

A variant of this measure is inter-quartile ratio, which can be defined as the 75th percentile (3rd quartile) income minus 25th percentile (1st quartile) income divided by the median (x^{q^2}) income.

11.2.3 Relative Standard Variation

The standard deviation divided by the mean can be used as one measure of dispersion. It is:

$$RSD = \frac{\sigma}{\mu} \quad (\text{RSD.1})$$

where σ and μ are standard deviation and mean of the distribution.

It can be equivalently defined as the standard deviation of relative incomes. Using definition of σ , one can find out that it lies in the interval of 0 and $(N-1)^{1/2}$, not in (0,1). The highest value depends on the size of distribution.

Since the measure uses all values, any transfer of income would be reflected in the measure. However, it should be noted that the measure is equi-sensitive to transfers at all levels. Whether a given amount d is transferred between $x_j = \text{Rs.}400$ and $x_k = \text{Rs.}500$, or between $x_j = \text{Rs.}10,000$ and $x_k = \text{Rs.}10,100$, the change in RSD is exactly the same.

We may finally note that the square of RSD is also quite often used as another measure of inequality, which is known as the coefficient of variance. Quite a few scholars suggest use of variance as a measure of inequality but we have not considered it here primarily because it is not unit-free. We think that an inequality measure must be unit-free.

11.2.4 Standard Deviation of Logarithms

One way of attaching greater importance to transfers at lower end (as required by Sen) is to consider some transformation of incomes. This transformation can easily be attained by considering the logarithms that stagger the income at lower levels.

This measure is defined in either of the following two ways:

$$SDL_1 = \left(\frac{1}{N} \sum_{i=1}^N (\log \mu - \log x_i)^2 \right)^{1/2} \quad (SDL.1)$$

$$SDL_2 = \left(\frac{1}{N} \sum_{i=1}^N (\log \hat{\mu} - \log x_i)^2 \right)^{1/2} \quad (SDL.2)$$

where μ and $\hat{\mu}$ are the arithmetic and geometric means respectively. While standard statistical literature prefers use of geometric mean the more common practice in literature on income inequality is one of using arithmetic means.

Cowell (1995) prefers to define these in terms of variance and calls the square of SDL_1 as the logarithm variance (V_1) and the square of SDL_2 as the variance of logarithms (V_2). Name of the second is clear from the expression but that of the first is derived from the fact that $(\log x - \log \bar{x})$ could be written as $\log (x/\bar{x})$. One can see that V_1 is equal to V_2 plus $\log (\hat{\mu} / \mu)$.

As these measures are in terms of ratios of incomes, any proportionate change in incomes would leave the magnitude of inequality unchanged when measured by these indices. But, unfortunately, a transfer from a richer person to a poorer person may raise the magnitude of inequality, particularly if the poorer person has income more than 2.72 times the mean of the distribution.

While the lower limit, irrespective of formula, is zero when everybody has the same income, the upper limit depends on the size of distribution and approaches infinity when N is large and when everybody except the richest, receives income equal to one unit (as zero is inadmissible in logarithmic transformation.) Further, if we face grouped data, it is convenient to use μ in place of $\hat{\mu}$ and μ_i in place of x_i .

The variance of logarithms is however decomposable. It is a property that is being given emphasis of late. It can be shown that V_2 is the sum of between-group component and within group component, latter being population-weighted sum of within-group V_2 's.

11.2.5 Champernowne Index

Champernowne (1973) makes use of the idea of geometric mean. It is a well known fact of an unequal distribution that its geometric mean is smaller than the arithmetic mean. The additive inverse of the ratio of geometric mean to arithmetic mean can duly be considered as an index of inequality. Formally, the index could be written as:

$$CII = 1 - \frac{\hat{\mu}}{\mu} \quad (CII.1)$$

where μ and $\hat{\mu}$ are, as stated earlier, arithmetic and geometric means of the income distribution. It is easy to see that its value is bound between 0 and 1.

One can obviously think of another measure where geometric mean is replaced by harmonic means.

These measures are sensitive to transfer to income and change is greater when the transfer takes place at lower end of the distribution. They are sensitive to transfer of income between two persons. One can try it by replacing x_j and x_k by $(x_j - d)$ and $(x_k + d)$ respectively and finding out the direction of the change. Or, one can use differential calculus.

The trouble with these indices is that they cannot be defined when any of the income is zero.

11.2.6 Hirschman-Herfindahl Indices

These indices were developed in the course of studying the commodity concentration in trade by Hirschman (1945) and in characterizing market monopoly in industry by Herfindahl (1950). Later, they were more used in capturing autonomy and dependence of units in a federation.

If each unit is a class in itself, $p_i = 1/N$, $i=1,2,\dots,N$. Then concentration could be captured through use of q_i 's. As the sum of q_i 's is always 1, Hirschman devised a measure which would capture the inequality among them. He proposed square root of the sum of squares of shares q_i $i=1,2,\dots,N$. That is,

$$H_1 = \left(N \sum_{i=1}^N q_i^2 \right)^{1/2} \quad (H.1)$$

which could be generalized as

$$H_1^* = \left(N \sum_{i=1}^N q_i^a \right)^{1/a}, a > 1 \quad (H.2)$$

Herfindahl devised a very similar measure, which has been more popular than the original (H.1). This is just the sum of share squares:

$$H_2 = \sum_{i=1}^N q_i^2 \quad (\text{H.3})$$

$$H_2^* = \sum_{i=1}^N q_i^a, a \geq 1. \quad (\text{H.4})$$

It is clear that, besides inequality among the shares, the value of these measures depends on N —the fewness or largeness of the number of units. For $N=2$, it has been suggested that $(1/N)$ could be subtracted from (H.3)

$$H_3 = \sum_{i=1}^N q_i^2 - \frac{1}{N} \quad (\text{H.5})$$

The minimum value of H_3 is zero. But it serves no great purpose. When $N=2$, for $q_1=0.99$ and $q_2=0.01$, while $H_2=0.98$, $H_3=0.48$. H_2 scores definitely better than H_3 in characterizing the scene of monopoly.

11.2.7 Kolm's Index

Let there be N incomes such that $N=nm$ where n is the number of different incomes and each income has m recipients. The number of equal pairs with a given income would be $m(m-1)/2$ and total number of equal pairs would be $n.m(m-1)/2$. Total number of all pairs would obviously be $N(N-1)/2 - nm(nm-1)/2$. One can think of an 'equality' index in terms of $nm(m-1)/nm(nm-1) = (m-1)/(N-1)$. The inequality index could then be constructed by subtracting it from 1: $1 - (m-1)/(N-1) = (N-m)/(N-1) = m(n-1)/(nm-1) = (nm-m)/(nm-1)$. In case, income x_i has f_i recipients, the measure is:

$$K = 1 - \frac{\sum f_i^2 - N}{N(N-1)} = \frac{N^2 - \sum f_i^2}{N(N-1)} \quad (\text{K.1})$$

The purpose of developing this curiosum due to Kolm (1996) is just to make one feel that there could be a variety of simple ways to approach the issue of measurement of inequality.

Check Your Progress 1

1) Define relative range measures of inequality. List out relative merits.

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2) How relative inter-quartile range is better than relative range?

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3) What is the relative mean deviation? If a transfer of income is between two persons both having income lower than the mean, will it change the magnitude of this index?

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4) Compare the two versions of standard logarithmic deviations.

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5) What is import of Champernowne Index?

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6) What is Hirfindahl index? What are its areas of application?

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